A TO Z MASTER GUIDE

Power BI A to Z

Power Query - Data Modeling - DAX - Dashboards - Service - Interview

Prepared specially for Koushik
Beginner-friendly A to Z edition

Inside: ETL | STAR SCHEMA | DAX | VISUALS | RLS | PROJECT

How to use	Oka topic chadivi code ni own-ga type cheyyi. First output guess cheyyi, tarvatha run cheyyi. Interview answer boxes ni loud-ga 2 times practice cheyyi.

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How to Use This Guide

Ee PDF Power BI ni zero nunchi interview level varaku nerchukovadaniki. Prathi main topic lo step, reason, expected result, common mistake and interview answer untayi. Menu names software version batti konchem change avvachu, kani workflow same untundi.

Step	What you do	Why
1. Observe	Concept and example chaduvu	Meaning ardham avutundi
2. Predict	Result mundu guess cheyyi	Logic build avutundi
3. Perform	Own-ga steps repeat cheyyi	Tool memory vastundi
4. Explain	Result enduku vachindo cheppu	Interview clarity vastundi
5. Practice	Two new questions solve cheyyi	Independent skill build avutundi

Interview answer formula	Definition one line -> process 2 or 3 points -> small example -> business use -> limitation. Ee order lo answer chepthe neat-ga and confident-ga untundi.

1. Power BI Foundation

1.1 What is Power BI?

Power BI is Microsoft's business intelligence platform for connecting, transforming, modeling, visualizing and sharing data. Power BI Desktop is mainly for authoring; the Power BI service supports collaboration, sharing, refresh and apps.

Easy Telugu	Raw data ni Power Query lo clean chesi, Model view lo tables connect chesi, DAX measures create chesi, Report view lo interactive visuals build chestam.

1.2 Main components

Component	Purpose
Power BI Desktop	Connect, transform, model and build reports
Power BI service	Publish, share, refresh, collaborate and distribute apps
Power BI Mobile	Consume optimized reports and dashboards
On-premises data gateway	Bridge service refresh/live queries to private sources
Power BI Report Server	On-premises report hosting in applicable environments

1.3 End-to-end workflow

Stage	Tool/view	Output
Connect	Home > Get data	Source query
Clean	Power Query Editor	Repeatable transformation steps
Model	Model view	Relationships and semantic model
Calculate	DAX	Measures/columns/tables
Visualize	Report view	Interactive report pages
Share	Publish to service	Workspace/report/app
Maintain	Refresh, gateway, security	Trusted current content

2. Desktop Interface and First Report

2.1 Views

View	Use
Report view	Build report pages and visuals
Table/Data view	Inspect loaded table data and calculated columns
Model view	Create/manage relationships and model properties
DAX query view (where available)	Explore and test DAX queries

2.2 First visual - step by step

Step	Action	Expected result
1	Home > Get data > Excel	Select workbook
2	Navigator -> choose Orders	Preview source
3	Choose Transform Data	Open Power Query for checking
4	Set types -> Close & Apply	Load model
5	Report view -> select Clustered column chart	Blank visual
6	Category to X-axis, Sales to Y-axis	Sales by category
7	Format pane -> title/labels/colors	Readable chart

Why Sales becomes Sum?	Numeric Sales field visual values area lo pedithe Power BI default summarization usually Sum. Column tools/model property lo summarization change cheyyachu; business meaning verify cheyyi.

3. Connect and Storage Modes

3.1 Common sources

- Files: Excel, CSV, Text, Folder, PDF where supported.
- Databases: SQL Server, PostgreSQL, MySQL, Oracle and others through connectors.
- Online: SharePoint, OneDrive, Web, Fabric and cloud services.
- Folder connector: combine monthly files with same schema.

3.2 Import, DirectQuery and live connection

Mode	How it works	Tradeoff
Import	Data copied into semantic model	Fast visuals; needs refresh and model size
DirectQuery	Queries source during interaction	Freshness/source security; latency and limitations
Live connection	Connect to existing semantic/analytical model	Central model; limited local modeling
Composite	Mix modes where supported	Flexibility with design complexity

Choice	Default-ga Import performance strong. Near real-time/security/source-size needs batti DirectQuery/composite consider cheyyi. Source capacity and user concurrency test cheyyi.

4. Power Query - Step by Step

4.1 Editor areas

Power BI Desktop Home > Transform data opens Power Query Editor. Left Queries pane, center preview, right Query Settings/Applied Steps, ribbon tabs Home/Transform/Add Column/View/Tools/Help untayi.

4.2 Clean sales data

Step	Action	Why/result
1	Choose correct header row	Meaningful field names
2	Set Date, Whole/Decimal, Text types	Correct storage and operations
3	Trim/Clean text columns	Remove hidden/extra spaces
4	Replace errors only after diagnosis	Controlled quality
5	Remove irrelevant columns	Smaller model
6	Filter invalid/test rows	Trusted analysis population
7	Rename query and steps	Maintainability
8	Close & Apply	Load transformed result

Applied Steps reason	Prathi click M transformation ga record avutundi. Refresh time lo original source meeda same ordered steps repeat avutayi. Previous step output next step input.

4.3 Data profiling

Power Query View tab lo Column quality, Column distribution and Column profile enable cheyyi. Profile row limit/status verify cheyyi. Errors, empty values, distinct/unique distribution inspect cheyyi.

4.4 Transform vs Add Column

Choice	Effect
Transform tab operation	Usually changes selected existing column
Add Column operation	Creates a new column and preserves original

When to preserve	Audit/validation kosam original field useful ayithe Add Column. Source field no longer needed and logic safe ayithe transform existing.

5. Merge, Append and Folder Combine

5.1 Merge queries

Step	Action	Reason
1	Home > Merge Queries	Choose primary and lookup tables
2	Select matching key(s) in same order	Define join
3	Choose join kind	Control matched/unmatched rows
4	Expand needed columns only	Avoid model bloat/duplicates
5	Check row count and unmatched keys	Validate cardinality

Row explosion	Lookup key unique kakapothe one fact row multiple lookup matches tho duplicate avutundi. Merge mundu lookup key uniqueness profile cheyyi.

5.2 Append queries

Home > Append Queries combines rows. Same meaning columns ki same name/type maintain cheyyi. Missing columns new result lo NULL values create chestayi.

5.3 Folder combine

Get data > Folder -> select folder -> Combine & Transform. Sample file transformation function all files ki apply avutundi. Schema drift and temporary files check cheyyi.

6. Data Modeling and Star Schema

6.1 Fact and dimension

Table	Contains	Example
Fact	Business events and numeric measures	One row per order line
Dimension	Descriptive attributes used to filter/group	Product, Customer, Date

Grain first	Fact table one row exactly emi represent chestundo first declare cheyyi. Example: one order item. Grain clear kakapothe totals double count avvachu.

6.2 Star schema

FactSales center lo; DimDate, DimProduct, DimCustomer, DimRegion one-to-many relationships tho connect. Dimension one side unique keys; fact many side foreign keys.

6.3 Relationship properties

Property	Meaning / guidance
Cardinality	One-to-many, one-to-one or many-to-many; match key uniqueness
Cross-filter direction	Prefer single dimension-to-fact unless a tested reason exists
Active	Default relationship used by filters and calculations
Inactive	Can be activated in a measure using USERELATIONSHIP

Ambiguous filters	Multiple bidirectional paths filter destination ki reach ayithe ambiguity/unexpected totals. Star schema and single direction model simpler and safer.

7. Date Table

7.1 Create with DAX

```
DateTable =
ADDCOLUMNS(
    CALENDAR(MIN(FactSales[Order Date]), MAX(FactSales[Order Date])),
    "Year", YEAR([Date]),
    "Month", FORMAT([Date], "MMMM"),
    "Month No", MONTH([Date]),
    "Year Month", FORMAT([Date], "YYYY-MM"),
    "Quarter", "Q" & FORMAT([Date], "Q")
)
```

Step	Why
Create continuous date table	Time functions need complete calendar
Mark as date table where appropriate	Identify canonical date column
Relate DateTable[Date] to FactSales[Order Date]	Date filters propagate
Sort Month by Month No	January to December, not alphabetic

Straight line / wrong months	Month name axis without Year can combine all years. Year-Month field use cheyyi and Month by Month No sort cheyyi. Date relationship active undo check cheyyi.

8. DAX Foundation

8.1 Measure vs calculated column

Calculated column	Measure
Computed row by row during refresh	Computed at query time for current filters
Stored in model	Definition stored; result depends on context
Useful for categories/relationships/sorting	Useful for aggregations and KPIs

```
Total Sales = SUM(FactSales[Sales])
Total Profit = SUM(FactSales[Profit])
Profit Margin = DIVIDE([Total Profit], [Total Sales], 0)
Order Count = DISTINCTCOUNT(FactSales[Order ID])
```

Why measure changes?	Region slicer select chesthe filter context FactSales rows ni reduce chestundi. Same Total Sales measure remaining rows SUM chestundi, so visual cell batti result change avutundi.

8.2 Row context and filter context

Calculated column/iterators row context provide current row. Visuals, slicers and CALCULATE create/modify filter context. CALCULATE can perform context transition.

8.3 CALCULATE

```
Technology Sales =
CALCULATE(
    [Total Sales],
    DimProduct[Category] = "Technology"
)
```

CALCULATE evaluates the expression under modified filters. Existing filters on the same column can be replaced unless KEEPFILTERS or other logic changes behavior.

9. Essential DAX Patterns

9.1 Iterators

```
Revenue =
SUMX(
    FactSales,
    FactSales[Quantity] * FactSales[Unit Price]
)
```

SUMX creates row context, calculates expression per row, then sums results. Plain SUM only sums one numeric column.

9.2 Filter and all

```
Sales % of All Categories =
DIVIDE(
    [Total Sales],
    CALCULATE([Total Sales], REMOVEFILTERS(DimProduct[Category])),
    0
)
```

Denominator reason	Current category numerator. REMOVEFILTERS category filter only remove chesi all-category total denominator. Other filters like Year/Region remain.

9.3 Selected value

```
Selected Region =
SELECTEDVALUE(DimRegion[Region], "Multiple / All")
```

9.4 Variables

```
Average Order Value =
VAR SalesAmount = [Total Sales]
VAR Orders = [Order Count]
RETURN
    DIVIDE(SalesAmount, Orders, 0)
```

Variables improve readability, avoid repeated evaluation and simplify debugging.

10. Time Intelligence

```
Sales YTD =
TOTALYTD([Total Sales], DateTable[Date])

Sales Previous Year =
CALCULATE([Total Sales], SAMEPERIODLASTYEAR(DateTable[Date]))

YoY Change = [Total Sales] - [Sales Previous Year]

YoY % = DIVIDE([YoY Change], [Sales Previous Year], 0)
```

Requirement	Why
Continuous date table	Correct period shifting
Date column relationship	Filters reach fact table
Correct granularity	Daily/monthly/yearly results reconcile
Fiscal calendar logic if needed	Business periods may differ from calendar

Blank previous year	Previous-year period source data lo lekapovachu, date table range incomplete, relationship inactive, or date field from fact used. Each check cheyyi.

11. Visuals and Field Wells

Question	Visual	Fields
Category comparison	Bar/column	Axis category, Values measure
Time trend	Line	X date, Y measure, optional legend
Part of whole	100% stacked/donut sparingly	Legend category, Values measure
Geography	Map	Location or lat/long, size/color measure
Relationship	Scatter	X/Y measures, Details entity
Drivers	Decomposition tree	Analyze measure, Explain by fields
Detail	Table/matrix	Rows/columns/measures
Single KPI	Card/KPI	Measure, optional target/trend

Axis problem	Visual type batti field wells names change avvachu. X-axis/Y-axis/Values/Legend locations current selected visual ni depend chestayi. First visual select cheyyi, then Build pane chudu.

11.1 Sort and data labels

Visual ellipsis (...) > Sort axis. Category charts descending measure sort business ranking ki useful. Data labels only when values readable; otherwise tooltip use cheyyi.

12. Filters, Slicers and Interactions

12.1 Filter scopes

Scope	Effect
Visual-level	Only selected visual
Page-level	All visuals on page
Report-level	All report pages
Drillthrough	Target page filtered by selected entity

12.2 Slicer

Add Slicer visual -> field add -> choose list/dropdown/between/relative date as applicable. View > Sync slicers can reuse selections across pages.

12.3 Edit interactions

Select a visual -> Format ribbon > Edit interactions -> for each other visual choose Filter, Highlight or None. Unexpected dashboard behavior often interaction settings valla vastundi.

Why card changed?	Slicer filter dimension nunchi relationship dwara fact rows ni filter chestundi; card measure remaining rows meeda recalculate chestundi.

13. Drill, Tooltips, Bookmarks and Buttons

13.1 Drill hierarchy

Create Date hierarchy Year > Quarter > Month or Location hierarchy Country > State > City. Visual header drill controls use chesi next level or expand all select cheyyi.

13.2 Drillthrough

Step	Action
1	Create detail page
2	Add Customer/Product field to Drill-through filters
3	Build detail visuals
4	Right-click a mark -> Drill through -> target page
5	Use auto back button

13.3 Report page tooltip

Create small page -> Page information Tooltip On -> Canvas settings Tooltip size -> build mini visuals -> assign tooltip page to source visual.

13.4 Bookmarks

View > Bookmarks and Selection panes. Set visual visibility/filter state, Add bookmark, configure Data/Display/Current page, then attach button action. Update bookmark after changes.

Bookmark issue	Bookmark Data option on unte slicer/filter states save chestundi and reset avvachu. Navigation-only bookmark ki Data off consider cheyyi.

14. Dashboard Design and Accessibility

14.1 Layout

- Top: 3-5 KPIs with units and comparison.
- Middle: primary trend and main breakdown.
- Bottom: supporting detail or drivers.
- Left/top compact filter area; avoid covering data.
- Use grid, alignment, grouping and consistent spacing.

14.2 Color and titles

One highlight color, neutral base and consistent semantic colors (profit positive, loss negative). Dynamic titles can use SELECTEDVALUE-based measures. Do not rely on color alone; labels/tooltips help accessibility.

14.3 Validation

Test	Expected
Grand total vs source	Matches after documented filters
Single category/region	All visuals respond as designed
No data selection	Friendly blank/zero message
Cross-year trend	Correct sorting and date relationship
Drill/bookmark/back	Navigation state behaves consistently

15. Power BI Service, Publish and Refresh

15.1 Key service objects

Object	Meaning
Workspace	Collaborative container for content
Semantic model	Modeled data used by reports
Report	Multi-page interactive visuals
Dashboard	Service-only single canvas of pinned tiles
App	Packaged distribution experience for consumers

15.2 Publish

Step	Action
1	Save PBIX and validate
2	Home > Publish
3	Choose authorized workspace
4	Open report in service
5	Configure credentials, refresh, permissions and app

15.3 Refresh and gateway

Cloud source credentials configure in service. On-premises/private source often needs configured gateway and data source mapping. Scheduled refresh availability depends on license/capacity and source.

Refresh fails	Credentials expired, gateway offline, source path/schema changed, privacy/firewall or timeout check cheyyi. Refresh history error details first chudu.

15.4 Incremental refresh

Large time-based tables ki RangeStart/RangeEnd parameters and refresh policy use chesi recent partitions only refresh cheyyachu. Query folding and source support verify cheyyi.

16. Row-Level Security

16.1 Static RLS

Step	Action
1	Modeling > Manage roles
2	Create role, select dimension table
3	Add rule such as [Region] = "South"
4	Modeling > View as to test
5	Publish and assign users/groups in service

16.2 Dynamic RLS

```
Security table columns: UserEmail, Region
Role filter:
Security[UserEmail] = USERPRINCIPALNAME()
```

Security table relationships filter allowed dimension/fact rows. RLS filters rows; it does not hide model objects such as columns or measures.

Security test	View as role + specific user test cheyyi. Bidirectional security paths and missing mappings verify cheyyi. Workspace permissions can affect how RLS applies.

17. Performance and Best Practices

- Use star schema and remove unused columns/rows.
- Prefer measures for aggregations; avoid unnecessary calculated columns.
- Use integer keys and appropriate data types.
- Avoid excessive bidirectional/many-to-many relationships.
- Reduce high-cardinality text columns where not needed.
- Use Performance Analyzer to identify slow visuals/DAX.
- Limit visuals per page and expensive custom visuals.
- Push foldable transformations toward the source where appropriate.
- Use variables and simpler DAX; validate filter removal scope.

Measure correctness first	Fast wrong answer use ledu. First grain, relationship, context correct cheyyi; tarvatha performance optimize cheyyi.

18. Complete Superstore Project

18.1 Requirements

- KPIs: Sales, Profit, Profit Margin, Orders, Quantity.
- Monthly trend with year filter.
- Region/Category/Sub-Category performance.
- Top customers and products.
- Map by state/city where data categories are correct.
- Drillthrough detail and reset filters bookmark.

18.2 Build order

Phase	Steps	Validation
Profile	Check rows, columns, types, nulls, keys	Document issues
Clean	Power Query types, names, invalid rows	Row count and totals
Model	Fact Orders + Date/Product/Customer/Location dims	One-to-many paths
DAX	Core measures and time intelligence	Table/matrix spot checks
Report	Overview, Product, Customer, Detail pages	Interactions and drill
Publish	Workspace, refresh, RLS, app	Test as consumer

18.3 Core measures

```
Total Sales = SUM(FactOrders[Sales])
Total Profit = SUM(FactOrders[Profit])
Profit Margin = DIVIDE([Total Profit], [Total Sales], 0)
Total Orders = DISTINCTCOUNT(FactOrders[Order ID])
Average Order Value = DIVIDE([Total Sales], [Total Orders], 0)
Sales PY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR(DateTable[Date]))
YoY % = DIVIDE([Total Sales] - [Sales PY], [Sales PY], 0)
```

18.4 Month sorting

DateTable Month name select -> Column tools > Sort by column > Month No. X-axis lo Year Month use chesthe years mix avvakunda chronological trend vastundi.

Final explanation	Dashboard lo 'what happened' KPI/trend, 'where' region/category, 'why' product/customer drivers, 'what next' recommendation show cheyyi.

19. Power BI Interview Questions with Model Answers

Q: 1. What is Power BI?

Answer: A business intelligence platform for connecting, transforming, modeling, visualizing and sharing data.

Q: 2. Desktop vs service?

Answer: Desktop is primarily for authoring models/reports. The service supports hosting, sharing, refresh, dashboards, apps and collaboration.

Q: 3. Report vs dashboard?

Answer: A report is multi-page and interactive. A Power BI dashboard is a service-only single canvas of tiles pinned from reports and other sources.

Q: 4. What is a semantic model?

Answer: The governed set of tables, relationships, calculations and metadata used by reports; it was formerly commonly called a dataset.

Q: 5. Power Query?

Answer: The data connection and transformation engine that records repeatable ETL steps in the M language.

Q: 6. DAX?

Answer: A formula language for calculations in tabular semantic models, including measures, calculated columns and calculated tables.

Q: 7. Measure vs calculated column?

Answer: A column is computed per row at refresh and stored. A measure is evaluated at query time under filter context.

Q: 8. Row context vs filter context?

Answer: Row context identifies a current row; filter context is the set of filters applied by visuals, slicers, relationships and DAX.

Q: 9. What does CALCULATE do?

Answer: It evaluates an expression in a modified filter context and can perform context transition.

Q: 10. SUM vs SUMX?

Answer: SUM totals one column. SUMX iterates a table, evaluates an expression per row and sums the results.

Q: 11. DIVIDE vs /?

Answer: DIVIDE safely handles zero/blank denominators and can return an alternate result.

Q: 12. RELATED?

Answer: Returns a value from the one-side related table under an existing many-to-one relationship, commonly in row context.

Q: 13. Star schema?

Answer: A central fact table connects to descriptive dimensions, improving model clarity, usability and often performance.

Q: 14. Fact vs dimension?

Answer: A fact table records events/measures at a declared grain; dimensions describe and filter those events.

Q: 15. Cardinality?

Answer: The key relationship pattern, such as one-to-many, determined by uniqueness on each side.

Q: 16. Single vs both filter direction?

Answer: Single usually propagates dimension filters to facts. Both propagates in two directions and can create ambiguity/security complexity.

Q: 17. Active vs inactive relationship?

Answer: The active relationship filters by default. An inactive one can be used in a measure with USERELATIONSHIP.

Q: 18. Many-to-many?

Answer: A relationship where neither side key is unique; use only with understood business grain and bridge-table alternatives.

Q: 19. Why date table?

Answer: It supplies continuous dates, calendar attributes, correct sorting and a reliable basis for time intelligence.

Q: 20. Why month sorts alphabetically?

Answer: Month name is text. Sort it by Month Number or use a real Year-Month date key.

Q: 21. Import vs DirectQuery?

Answer: Import stores compressed data for fast analysis and needs refresh. DirectQuery sends queries to the source and depends on source performance.

Q: 22. Merge vs append?

Answer: Merge joins columns by key; append stacks rows.

Q: 23. Query folding?

Answer: Power Query pushes compatible transformations back to the source so the source performs them, often improving efficiency.

Q: 24. What are Applied Steps?

Answer: The ordered transformations executed on source data during refresh.

Q: 25. RLS?

Answer: Row-level security filters model rows for assigned users/roles; it does not hide columns, tables or measures.

Q: 26. Static vs dynamic RLS?

Answer: Static roles use fixed rules. Dynamic RLS maps the current user identity to allowed entities, often through a security table.

Q: 27. Gateway?

Answer: A managed bridge allowing the cloud service to access supported on-premises/private data sources.

Q: 28. Scheduled refresh?

Answer: The service reprocesses source data on configured times subject to credentials, gateway, source and license/capacity limits.

Q: 29. Incremental refresh?

Answer: Partitions a time-based table so only recent or changed periods refresh according to policy.

Q: 30. Drill down vs drillthrough?

Answer: Drill down moves through a hierarchy in the same visual. Drillthrough navigates to a filtered detail page.

Q: 31. Bookmark?

Answer: A saved report state controlling page, filters and display selections depending on configured options.

Q: 32. Edit interactions?

Answer: Controls whether one visual filters, highlights or does not affect another visual.

Q: 33. Tooltip page?

Answer: A specially sized report page displayed when hovering over a mark, providing contextual detail.

Q: 34. Filter scopes?

Answer: Visual, page and report filters affect progressively broader report areas.

Q: 35. ALL vs REMOVEFILTERS?

Answer: Both can remove filters in calculations; REMOVEFILTERS clearly expresses filter removal, while ALL also returns a table in relevant contexts.

Q: 36. SELECTEDVALUE?

Answer: Returns the single visible value, otherwise an alternate result.

Q: 37. What is context transition?

Answer: CALCULATE converts an existing row context into equivalent filter context for its expression.

Q: 38. Why totals look wrong?

Answer: Often the measure is recalculated at total context rather than adding displayed rows; verify business logic, grain and iterator requirements.

Q: 39. Why blank visual?

Answer: Check filters, relationships, missing keys, DAX blank behavior, data types and whether source rows exist.

Q: 40. How to optimize a model?

Answer: Use star schema, remove unused/high-cardinality columns, correct types, efficient measures, fewer visuals and inspect Performance Analyzer.

Q: 41. What is Performance Analyzer?

Answer: A Desktop tool recording visual query/display timings to identify slow report elements.

Q: 42. What is a workspace?

Answer: A service collaboration container holding semantic models, reports, dashboards and related content.

Q: 43. App?

Answer: A packaged, controlled distribution experience built from workspace content for consumers.

Q: 44. How do you validate a report?

Answer: Reconcile totals with source, test filters/edge cases, verify grain/relationships, inspect blanks and test user security/navigation.

Q: 45. How do you explain a Power BI project?

Answer: State the business problem, sources, cleaning, model grain, key DAX, dashboard decisions, validation, deployment and impact.

20. Practical Interview Tasks

Task	Strong answer
Build sales dashboard	Requirement -> clean -> model -> DAX -> report -> validate
Fix straight month line	Date grain, Year-Month, sort, filters and relationship
Wrong grand total	Explain evaluation context and rewrite measure if needed
Secure region users	Security mapping, dynamic RLS, View As and service assignment
Slow report	Performance Analyzer, model size, DAX, visual count and source
Refresh failure	History, credentials, gateway, schema/path and folding/timeouts

Answer style	Clicks matrame cheppaku. Data grain, relationship, filter context and validation enduku important-o cheppu. Adhe strong Power BI interview answer.

21. 30-Day Power BI Plan

Days	Topics	Output
1-3	Interface, connect, basic visuals	First report
4-7	Power Query types/cleaning	Clean refreshable query
8-9	Merge, append, folder	Combined dataset
10-13	Star schema and relationships	Validated model
14-18	DAX measures and context	Measure library
19-20	Date table/time intelligence	Trend page
21-23	Visuals, filters, interactions	Overview page
24-25	Drill, tooltips, bookmarks	Interactive UX
26	Service, refresh, RLS	Deployment checklist
27-28	Superstore project	Portfolio PBIX
29-30	Interview and mock task	45 answers + timed build

Official References and Version Note

Software interfaces update avutayi. Ee guide concepts and click paths official documentation tho verify chesanu. Option exact place kanipinchakapothe ribbon search/help use cheyyi.

Official source	URL
Power BI documentation	https://learn.microsoft.com/en-us/power-bi/
Power Query in Desktop	https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-query-overview
Power Query applied steps	https://learn.microsoft.com/en-us/power-query/applied-steps
Star schema guidance	https://learn.microsoft.com/en-us/power-bi/guidance/star-schema
Relationships guidance	https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-relationships-understand
RLS guidance	https://learn.microsoft.com/en-us/power-bi/guidance/rls-guidance
Power BI service concepts	https://learn.microsoft.com/en-us/power-bi/fundamentals/service-basic-concepts